

Linyang Power Atlantic 5MWh BESS RTE Calculation

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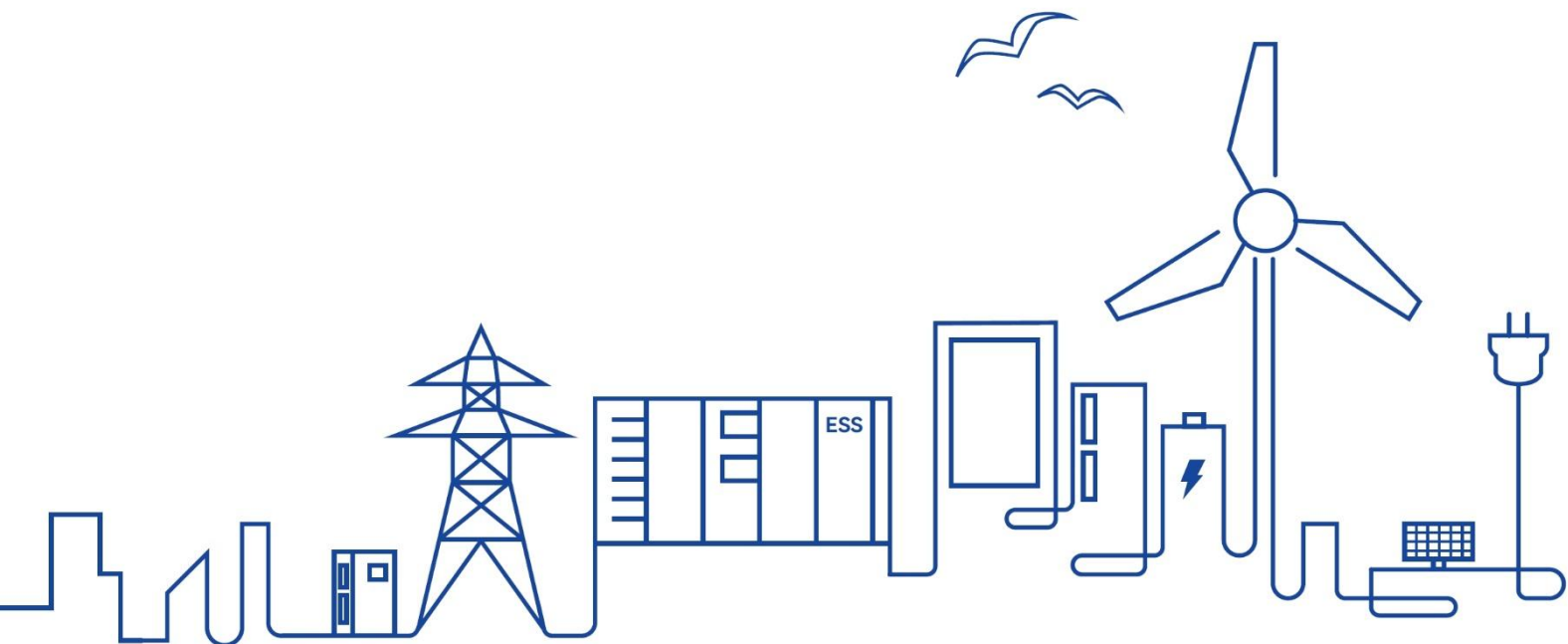


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● Abbreviations

Aux	Auxiliary Power Consumption
BESS	Battery Energy Storage System
COD	Commercial Operation Date
DOD	Depth of Discharge
POC	Point of Connection
RTE	Rouond-trip Efficiency
SOH	State of Health

1. BESS RTE Calculation Description

The Round-Trip Efficiency (RTE) of a Battery Energy Storage System (BESS) refers to the efficiency of the system in charging and discharging energy. It indicates the percentage of energy that can be recovered during discharge compared to the energy input during charging.

The RTE calculation for a BESS typically involves determining the energy efficiency of the system by comparing the energy discharged during the discharging process to the energy charged during the charging process. The formula for calculating RTE for a BESS is:

$$RTE = \frac{\text{Energy Discharged}}{\text{Energy Charged}} \times 100\%$$

In this formula:

- ♦ *Energy Discharged* is the actual energy discharged by the BESS during the discharging process.
- ♦ *Energy Charged* is the actual energy charged into the BESS during the charging process.

2. RTE Calculation Model

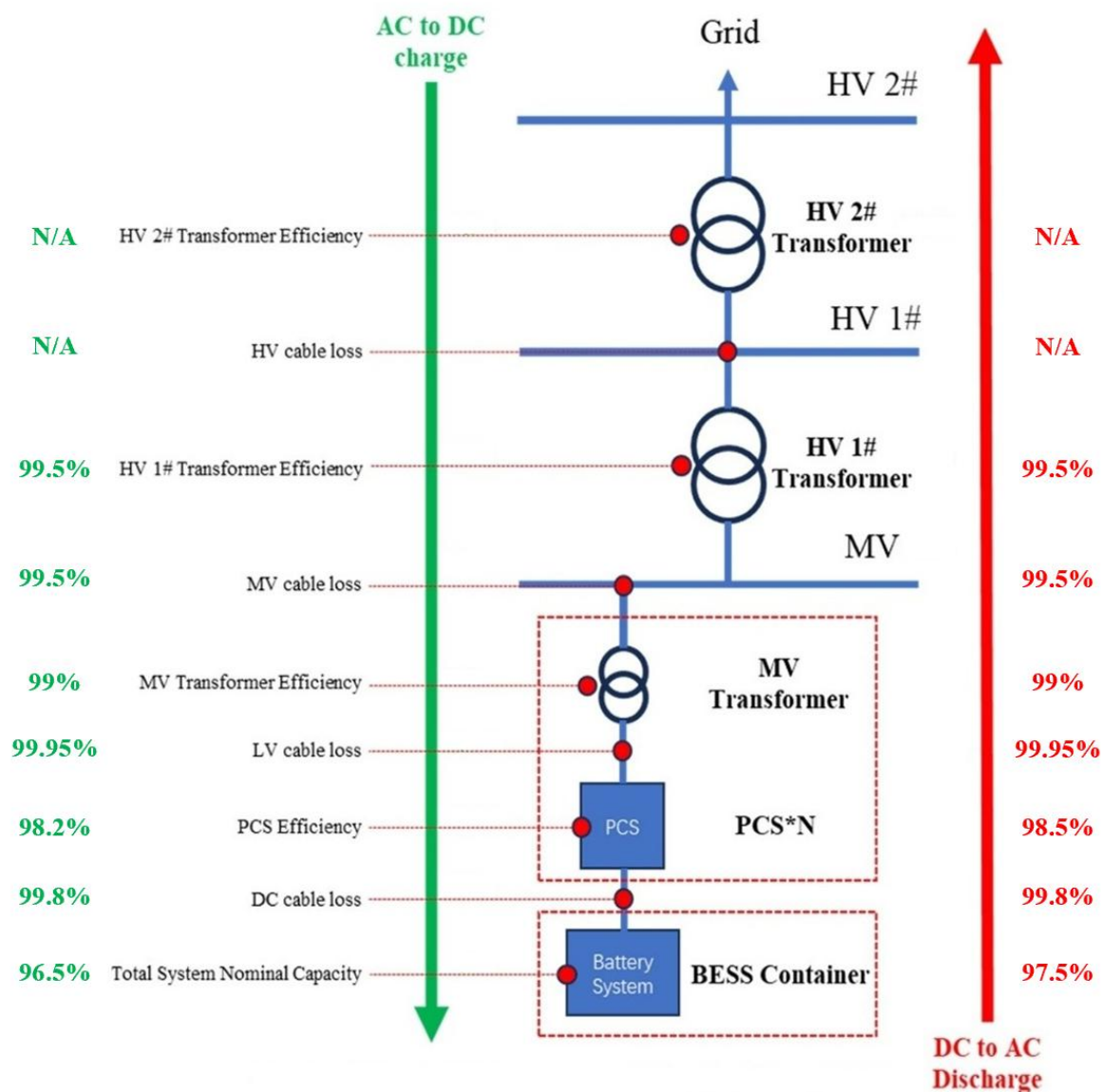


Fig.1 RTE Calculation Model (without Auxiliary Power)

The calculation of round-trip efficiency is as follows:

$$\begin{aligned} RTE &= 99.95\%(HV \text{ transformer efficiency}) \times 99.95\%(MV \text{ cable loss}) \\ &\times 99\%(MV \text{ transformer efficiency}) \times 99.95\%(LV \text{ cable loss}) \\ &\times 98.2\%(PCS \text{ rectifying efficiency}) \times 99.8\%(DC \text{ cable loss}) \\ &\times 96.5\%(Battery \text{ charging efficiency}) \times 97.5\%(Battery \text{ discharging efficiency}) \\ &\times 99.8\%(DC \text{ cable loss}) \times 98.5\%(PCS \text{ inverting efficiency}) \\ &\times 99.95\%(LV \text{ cable loss}) \times 99\%(MV \text{ transformer efficiency}) \\ &\times 99.5\%(MV \text{ cable loss}) \times 99.5\%(HV \text{ transformer efficiency}) \\ &\approx 87.8\% \end{aligned}$$